Course Project Phase IV (Final Deliverables)

Comprehensive IEEE Research Report

Objective: The primary objective of this final deliverable is to produce a comprehensive, academic-grade research report covering all aspects of your blockchain project.

Formatting & Submission Rules:

- **Length:** The final report must be a minimum of **7 pages** (excluding references).
- **Format:** The document must strictly adhere to the double-column **IEEE Conference Format**.
- **Tooling:** You are required to write and format this paper using LaTeX via Overleaf.
- **Template Link:** Use this exact template to start your paper: IEEE Conference Template on Overleaf

Your final IEEE research paper must include the following sections:

1. Title & Authors

- The exact title of your research project.
- The names, roll numbers, and institutional affiliations of all group members.

2. Abstract

- A concise, one-paragraph summary (approx. 150-250 words) describing the core motivation, the specific blockchain/cryptographic problem addressed, the experiments conducted, and the primary quantitative findings.

3. Introduction

- **Background and Motivation:** Clearly articulate the background of the research topic. Outline the motivations behind the chosen area, emphasizing its relevance to the current Web3/blockchain ecosystem.
- **Problem Statement:** Define the exact technical problem, vulnerability, or inefficiency your research addresses. Highlight the gap in existing knowledge or current protocols.
- **Contribution:** Clearly state your intended contribution. How does your proposed architecture, algorithm, or protocol advance current understanding or provide novel insights?

4. Related Literature

- **Identification of Key Works:** Summarize 3-5 foundational or recent research papers directly relevant to your topic.
- **Critical Analysis:** Discuss the strengths, weaknesses, and limitations of these existing solutions (e.g., *Why did previous ZK-rollups fail here? Why is current SHA-512 insufficient for your modeled scenario?*). Articulate exactly how your project builds upon or diverges from them.

5. Proposed Methodology

- Explain your core theoretical solution. What is mathematically or architecturally new about your idea?
- Detail your system design, cryptographic assumptions, or consensus logic. Explain *why* you believe this methodology is superior to previous attempts (e.g., better security guarantees, lower computational overhead, enhanced privacy).

6. Implementation of Proposed Methodology

- Describe the exact engineering of your proposed idea so that another blockchain researcher could reproduce your setup.
- **Technical Details:** Include details on the frameworks used (e.g., Rust, Go, Circom, Solidity), the division between on-chain and off-chain architecture, and specific algorithmic logic. Explain the reasoning behind your core design decisions.

7. Experimental Design

- Describe the rigorous testing and analyses you conducted to validate your contribution.
- **Evaluation Metrics:** Explicitly state the metrics you measured and *why*. For blockchain projects, this must include relevant data points such as:
    - Gas consumption / Transaction fees
    - Transactions Per Second (TPS) / Throughput
    - Proof generation/verification time (for ZK projects)
    - CPU/Memory overhead or hash rates (for cryptographic algorithms)

8. Experimental Results & Discussion

- **Data Visualization:** Present your findings using clear tables, charts, and graphs.
- **Analysis:** Discuss the observed trends. Did the system perform as theoretically expected? Provide insights into why your results turned out the way they did.
- **Limitations & Future Work:** Include at least one paragraph explaining what technical questions remain unanswered by your experiments and the natural next steps for this direction of research.

9. Conclusion

- Summarize the main takeaway points from your work in one or two paragraphs.
- **Ethical Implications:** Add a final paragraph discussing the broader impact of your project (e.g., privacy enhancements, financial fairness, regulatory compliance, or decentralization risks).

10. References

- All sources, including research papers, whitepapers, and official protocol documentation, must be properly cited using strict **IEEE citation standards**.

11. Appendix (Codebase Repository)

- Provide a public link (e.g., GitHub) to your finalized codebase. The repository must include a `README.md` with instructions on how to compile and run your code.